



TOSHKENT SHAHRIDAGI INHA UNIVERSITETI

INHA UNIVERSITY IN TASHKENT

STUDENT NAME:

STUDENT ID NUMBER:

SECTION NUMBER IN THIS COURSE:

FINAL EXAMINATION – SPRING 2024

COURSE NAME:

Computer Architecture

COURSE NUMBER:

SOC2060

EXAMINATION DATE:

TIME:

EXAMINATION DURATION:

60 Minutes

ADDITIONAL MATERIALS
ALLOWED TO USE:

None

SPECIAL INSTRUCTIONS:

- Answers will only be evaluated if they are readable.
- Show all your work. For some questions, you may get partial credit even if the end result is wrong due to a calculation mistake. If you make assumptions, state your assumptions clearly and precisely.

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST:

Desks should be free from all unnecessary items (books, notes, technology, food, water, clothes)

Use of any electronic device (Phone, iPod, iPad, laptop) is not allowed during the examination.

Cheating, talking to fellow students, singing, turning back are not allowed

Write your Name (*capital letters*), ID number and Section number in each page of your examination paper.

Final answers must be written by **only blue or black, non-erasable pen**. Do not use highlighters or correction pen.

All answers should be written in the space provided for each question, unless specified the other way.

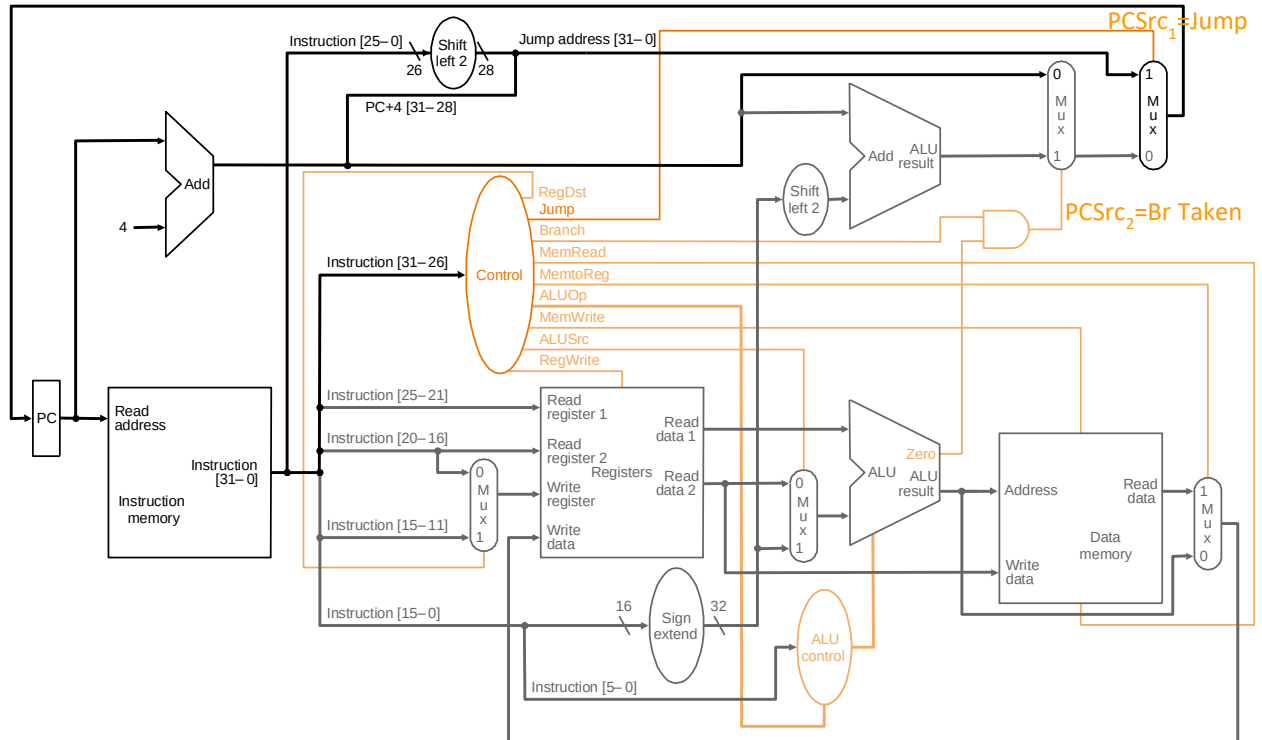
If you have a problem please raise your hand and wait quietly for a Proctor.

You are not allowed to leave the exam room until you submit the exam papers.

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(a)[10 points] You are about to execute J-type “j” (jump) instruction. List all the values of control signals to process “j”. If a control signal doesn’t affect the execution of “j”, write X meaning “don’t care”.

Control Signal	Value
RegDst	X
PCSrc ₁	1
Branch	0
MemRead	0
MemtoReg	X
ALUControl _{2:0}	XXX
MemWrite	0
ALUSrc	X
RegWrite	0
PCSrc ₂	0

(b)[10 points] You are about to execute I-type “subi” (Subtract Immediate) instruction. List all the values of control signals to process I-type “subi” instruction. If a control signal doesn’t affect the execution of “subi”, write X meaning “don’t care”.

Control Signal	Value
RegDst	0
PCSrc ₁	0
Branch	0
MemRead	0
MemtoReg	0
ALUControl _{2:0}	110
MemWrite	0
ALUSrc	1
RegWrite	1
PCSrc ₂	0

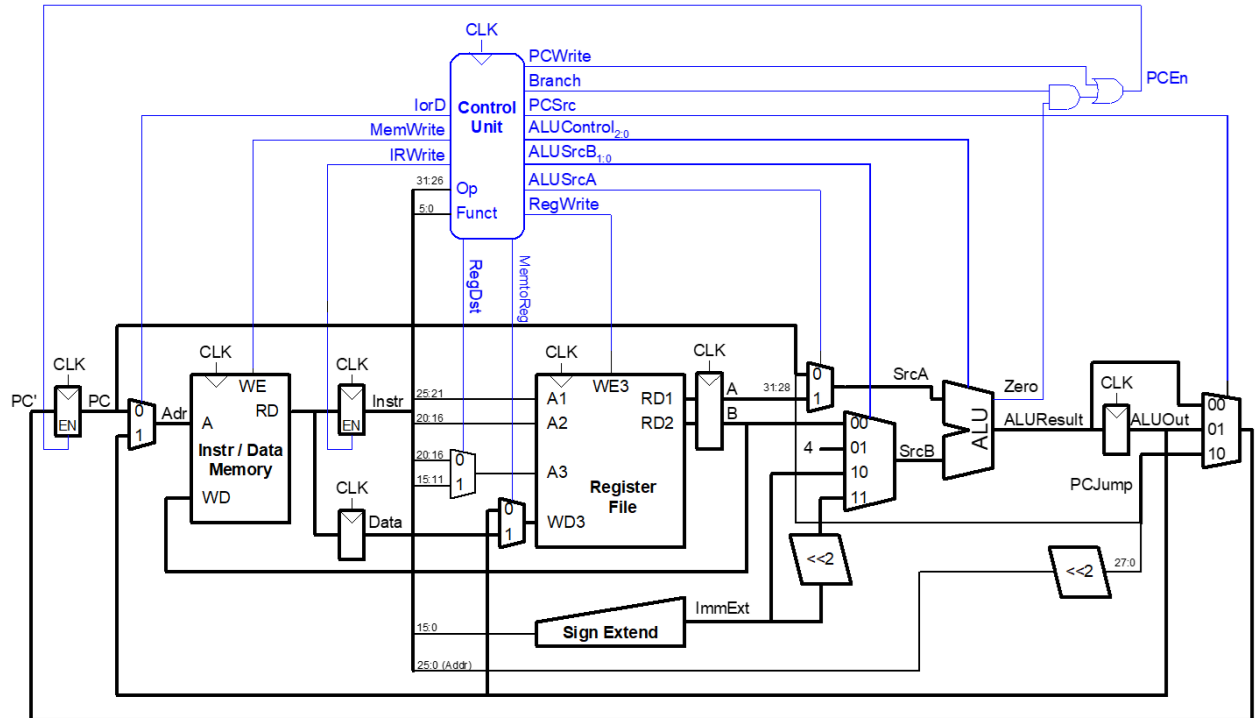
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3.[40 points] Multi-Cycle Processor Datapath

Below is the picture of a multi-cycle MIPS processor. And ALU opcodes and instruction formats are same as question 2.



ALUControl_{2:0} is also the same as question 2.

(a)[10 points] List all the values of control signals at “ID/RF” (Decode/Register Operand Fetch) phase of J-type “j” (Jump). If a control signal doesn’t affect the “ID/RF”, write X meaning “don’t care”.

Control Signal	Value
IorD	X
RegWrite	0
IRWrite	0
RegDst	X
MemtoReg	X
PCWrite	1
PCSrc	10
ALUControl _{2:0}	XXX
ALUSrcB _{1:0}	XX
ALUSrcA	X

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(b)[10 points] List all the values of control signals at “ID/RF” (Instruction Decode/Register Operand Fetch) phase of “beq” (Branch on Equal). If a control signal doesn’t affect the “ID/RF”, write X meaning “don’t care”.

Control Signal	Value
lorD	X
RegWrite	0
IRWrite	0
RegDst	X
MemtoReg	X
PCWrite	0
PCSrc	XX
ALUControl _{2:0}	010
ALUSrcB _{1:0}	11
ALUSrcA	0

(c)[10 points] List all the values of control signals at “EX/AG” (EXecute/Address Generation) phase of “addi” (Add Immediate). If a control signal doesn’t affect the “EX/AG”, write X meaning “don’t care”.

Control Signal	Value
lorD	X
RegWrite	0
IRWrite	0
RegDst	X
MemtoReg	X
PCWrite	0
PCSrc	XX
ALUControl _{2:0}	010
ALUSrcB _{1:0}	10
ALUSrcA	1

(d)[10 points] List all the values of control signals at “WB” (Write Back) phase of I-type “lw” (Load Word). If a control signal doesn’t affect the “WB”, write X meaning “don’t care”.

Control Signal	Value
lorD	X
RegWrite	1
IRWrite	0
RegDst	0
MemtoReg	1
PCWrite	0
PCSrc	XX
ALUControl _{2:0}	XXX
ALUSrcB _{1:0}	XX
ALUSrcA	X

4.[30 points] Caching

We have a byte addressable machine with L1, L2 cache and a memory. The size of a block is 32 bytes. The size of L1 cache is 128 bytes, the size of L2 cache is 2048 bytes, and the size of the memory is 2 giga bytes. The latency of L1 cache is 5 cycles, the latency of L2 cache is 10 cycles, and the latency of memory is 100 cycles. The L1 cache is a direct-mapped cache and the L2 cache is a fully associative cache. You are

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running a program which reads a byte from the memory 5 times. Assume both caches are empty, and the memory has all the data which your instructions are accessing.

(a)[6 points]

The address of the first byte to load is 193 and now the byte is loaded into a register. Specify how many cycles are needed for the load and specify the starting address(es) of block(s) of L1 and L2 caches.

The number of cycles needed: 115

Starting address(es) of block(s) in L1 cache: 192

Starting address(es) of block(s) in L2 cache: 192

(b)[6 points]

The address of the next byte to load is 175 and now the byte is loaded into a register. Specify how many cycles are needed for the load and specify the starting address(es) of block(s) of L1 and L2 caches.

The number of cycles needed: 115

Starting address(es) of block(s) in L1 cache: 192, 160

Starting address(es) of block(s) in L2 cache: 192, 160

(c)[6 points]

The address of the next byte to load is 223 and now the byte is loaded into a register. Specify how many cycles are needed for the load and specify the starting address(es) of block(s) of L1 and L2 caches.

The number of cycles needed: 5

Starting address(es) of block(s) in L1 cache: 192, 160

Starting address(es) of block(s) in L2 cache: 192, 160

(d)[6 points]

The address of the next byte to load is 450 and now the byte is loaded into a register. Specify how many cycles are needed for the load and specify the starting address(es) of block(s) of L1 and L2 caches.

The number of cycles needed: 115

Starting address(es) of block(s) in L1 cache: 160, 448

Starting address(es) of block(s) in L2 cache: 192, 160, 448

(e)[6 points]

The address of the next byte to load is 199 and now the byte is loaded into a register. Specify how many cycles are needed for the load and specify the starting address(es) of block(s) of L1 and L2 caches.

The number of cycles needed: 15

Starting address(es) of block(s) in L1 cache: 192, 160

Starting address(es) of block(s) in L2 cache: 192, 160, 448